

gRNA Predictor

AI-Powered CRISPR Guide RNA Designer with Gaussian Proximity Ranking

r = 0.640

Kim 2019 independent
(n=1,828 novel guides)

r = 0.537

Doench held-out
(20%, n=938)

450-dim

Feature vector
(XGBoost, 500 trees)

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Unit tests
passing

Independent Validation Results

Dataset	n	Spearman r	Notes
Kim 2019 novel-only ★	1,828	0.640	0% Doench overlap — primary benchmark
Doench 2016 held-out	938	0.537	Our 20% held-out (honest)
Azimuth (same split)	938	0.654	Asymmetric: Azimuth trained on 100% Doench
Chari 2015 (293T)	10	0.770	CI $\approx \pm 0.74$ — treat as supplementary
Chari 2015 (K562)	10	0.758	CI $\approx \pm 0.74$
Xu 2015 (human)	35	0.424	

CRISPRscan (zebrafish) 1,020 0.081 Expected low — trained on human

Novel contribution: Gaussian proximity-weighted ranking

$\text{combined_score} = (1-w) \times \text{efficiency} + w \times \exp(-d^2/2\sigma^2)$ where $\sigma = 50$ bp, $w \in [0,1]$ (user-tunable).

No existing tool (Azimuth, CRISPRscan, CRISPOR) automates this efficiency-proximity tradeoff.

gRNA Predictor — Publication Figures

Fig 1 — Predicted vs Experimental Efficiency

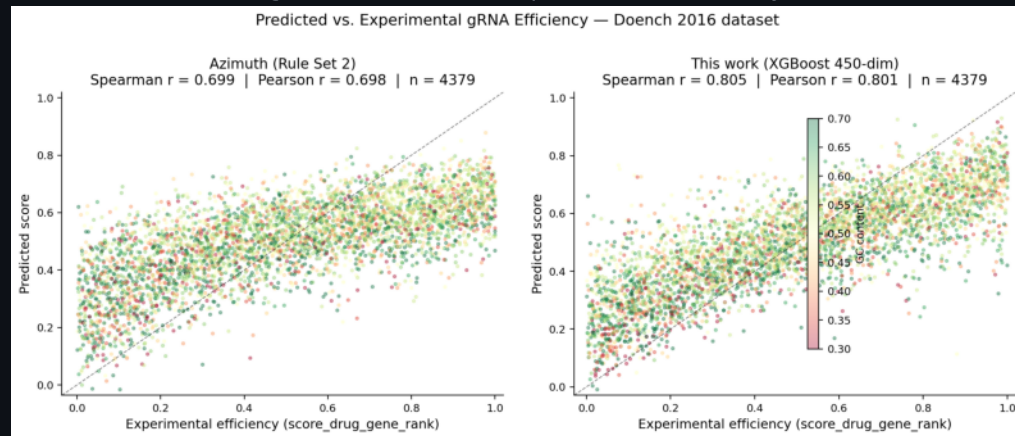


Fig 4 — Feature Ablation (SHAP)

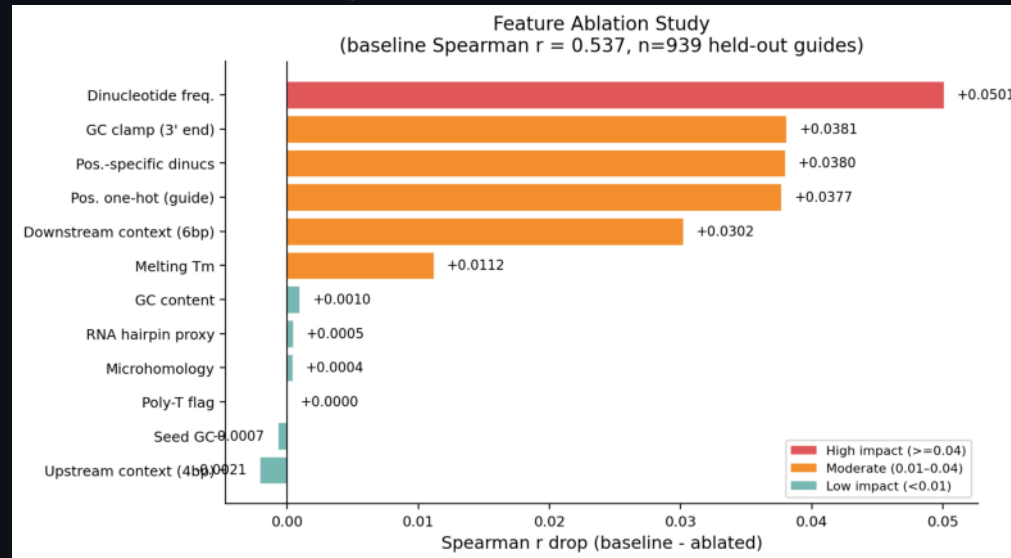


Fig 5 — Benchmark Comparison Heatmap

